Experiment No:3 Energy and Power Analysis of Signals

Objective:

(i) To calculate the energy and average power of given continuous-time and discrete time signals and classify them as energy signals, power signals, or neither based on their energy and power characteristics.

Procedure:

1. Define the time vector and generate the selected signals.

2. Compute the squared magnitude of each signal.

3. Numerically integrate over a sufficiently large interval for energy. 4. Compute average power over an interval containing several periods for the periodic signal.

5. Compare numerical results with analytical expectations.

6. Classify the signals and record observations.

Matlab Coding:

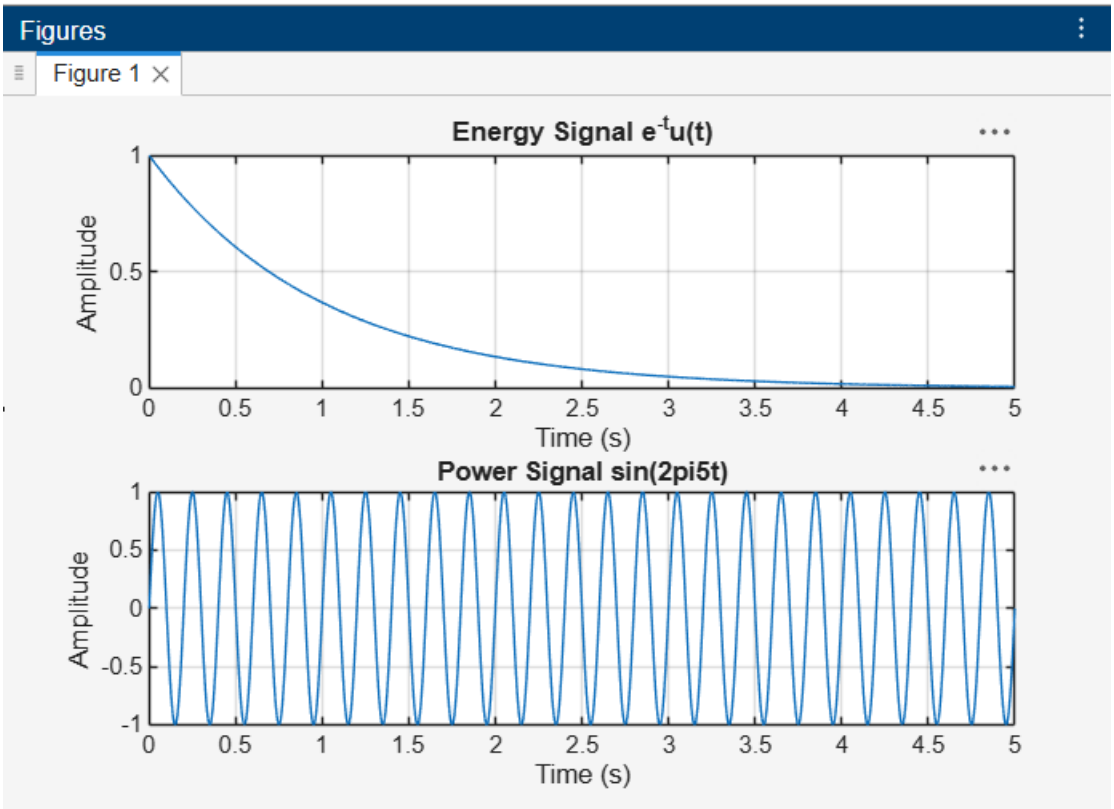
Editor

exp2aSS.m | exp2bSS.m | exp2cSS.m | exp2dSS.m | exp3SS.m

/MATLAB Drive/exp3SS.m

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% Experiment 3: Energy and Power Analysis
clc; clear; close all;
dt = 0.001;
t = 0:dt:5;
xE = exp(-t);
xP = sin(2*pi*5*t);
E_num = trapz(t,abs(xE).^2);
T0 = 1/5;
idx = t <= 2;
P_num = (1/(t(find(idx,1,'last'))-t(1))) * ...
    trapz(t(idx),abs(xP(idx)).^2);
figure;
subplot(2,1,1);
plot(t,xE); grid on;
title('Energy Signal e^{-t}u(t)');
xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,1,2);
plot(t,xP); grid on;
title('Power Signal sin(2pi5t)');
xlabel('Time (s)'); ylabel('Amplitude');
```

Output:



Result:

The numerical energy of $x_1(t) = e^{(-t)}\, u(t)$ is approximately 0.5000, and the numerical average power of $x_2(t) = \sin(2\pi 5t)$ is approximately 0.5000. The results agree with the theoretical values within the expected numerical accuracy.